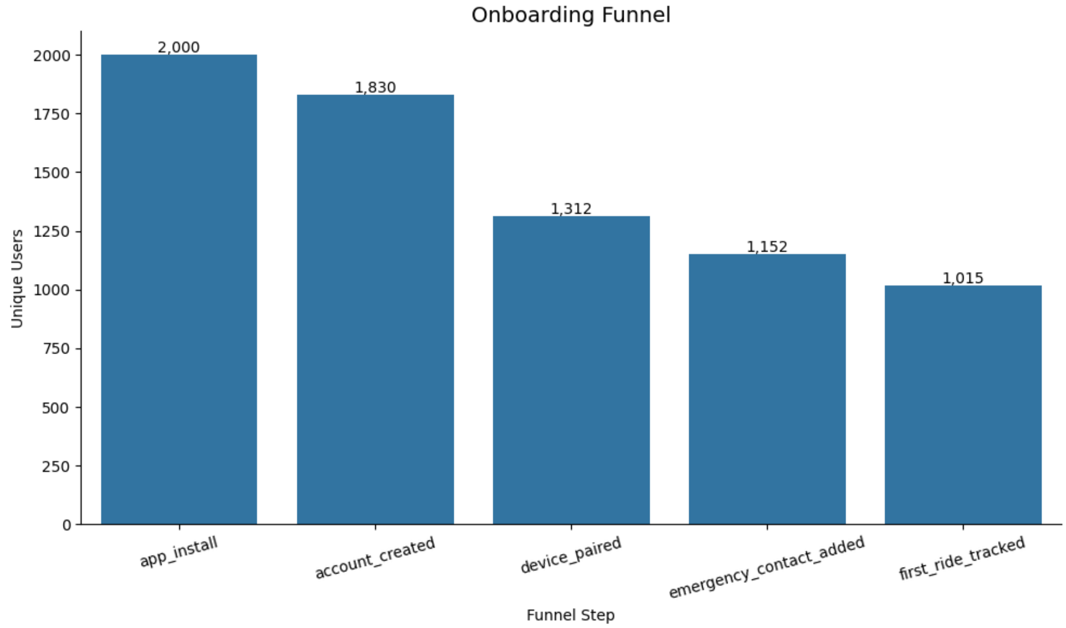
Task 1 - Funnel & Retention Analysis

Objective

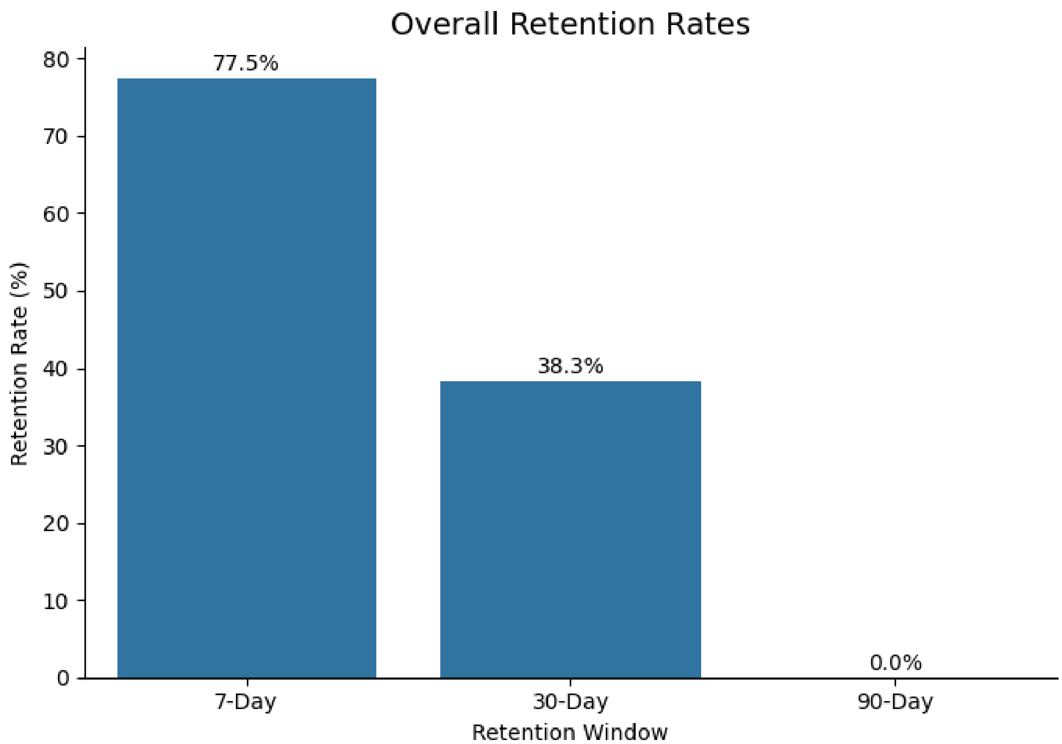
The goal of this analysis was to understand where users drop off during onboarding, how retention changes across acquisition cohorts, and which early product behaviours are most associated with long-term engagement.

Onboarding Funnel Analysis

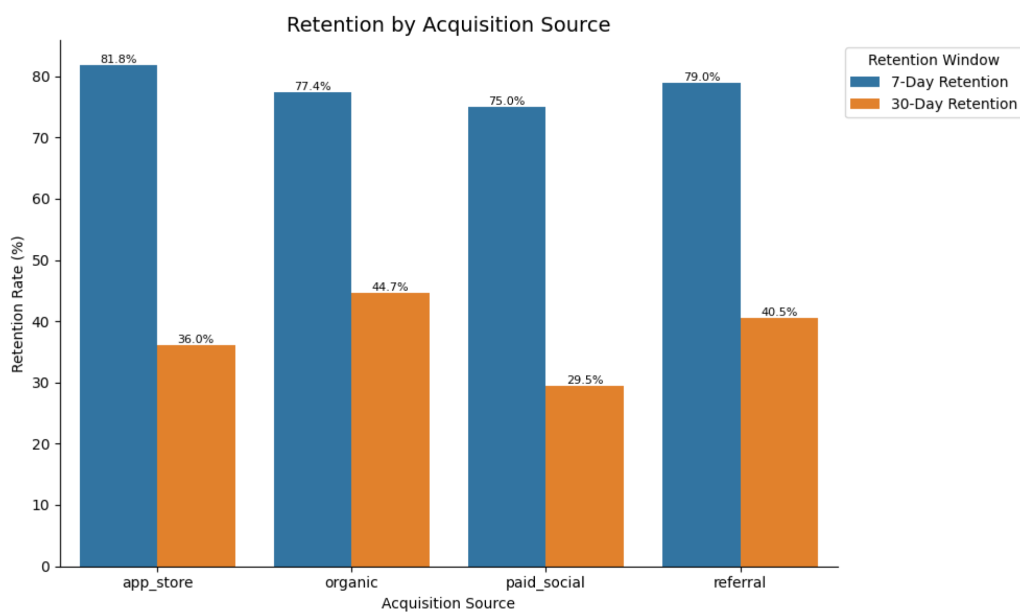


The onboarding funnel showed that 1,830 out of 2,000 users (~92%) progressed beyond app install and successfully created an account. The largest drop-off occurred during device pairing, where users declined from 1,830 to 1,312 (~28% drop), suggesting that setup friction may be preventing users from reaching Quin’s core product experience. A second meaningful drop-off occurred before first_ride_tracked, with only 1,015 users completing a tracked ride. This means that only ~51% of installed users reached the product’s primary activation moment. Based on these findings, we can think about improving the device pairing experience, introducing stronger onboarding guidance toward first ride completion, and testing early lifecycle engagement nudges during the first week after signup.

Retention Analysis (D7/D30/D90)



Overall retention declined from 77.5% on Day 7 to 38.3% on Day 30, suggesting that while many users initially engage with the product, a smaller group continues using it consistently over time. This indicates an opportunity to improve habit formation and repeat engagement after onboarding. Since the dataset only covers an 89-day observation window, 90-day retention could not be evaluated reliably for newer signup cohorts.



Retention rates also varied across acquisition cohorts. Organic users showed the strongest 30-day retention (~44.7%), followed by referral users (~40.5%), while paid social users retained at a lower rate (~29.5%). This suggests that acquisition quality may differ by channel, with organic and referral cohorts potentially bringing in users with stronger initial intent or better product fit. Based on these findings, we can consider prioritizing stronger onboarding and retention interventions for lower-retention paid cohorts.

Leading Indicators of Retention

To identify early behavioural signals associated with retention, I analyzed user actions occurring within the first 7 days after signup and compared them against 30-day retention outcomes.

The strongest behavioural indicators of retention were:

- ride_tracked
- first_ride_tracked
- push_notification_tapped

Users who tracked a ride within their first week retained at substantially higher rates than the overall user base, suggesting that reaching Quin's core product experience early is strongly associated with ongoing engagement. Similarly, users who completed their first tracked ride or interacted with push notifications also showed meaningfully stronger retention outcomes.

Overall, the analysis suggests that getting users to complete riding activity early in their lifecycle can be one of the highest-leverage opportunities for improving long-term retention.

Task 2 - Experiment Design

Objective

The objective of this experiment is to reduce onboarding drop-off and improve early user activation by increasing the percentage of users who complete first_ride_tracked within 7 days of signup.

Task 1 analysis showed that users who completed ride-tracking behaviors had substantially higher long-term retention compared to users who did not. However, a large proportion of users appeared to drop off before reaching their first tracked ride. I also reviewed the onboarding experience within App, which revealed several friction points, including a lengthy sequential setup process, SMS verification requirement, and limited onboarding progress visibility before user could experience Quin's core product value.

This suggests onboarding friction may be preventing users from reaching the primary product experience where they can begin tracking rides with active crash-detection protection.

Hypothesis

If Quin streamlines onboarding by simplifying setup flows, improving onboarding progress visibility, and reducing friction before users reach the primary application experience, then the percentage of users who complete first_ride_tracked within 7 days will increase, because users will experience lower onboarding fatigue and reach Quin's core product value faster.

Experiment Design

Population: The experiment population will consist of newly registered Quin users during the experiment period. Users will be randomly assigned at the user level into Control and Treatment groups using a 50/50 split.

Control Group (Variant A)

Users in the control group will experience Quin's existing onboarding flow and activation experience.

Treatment Group (Variant B)

Users in the treatment group will experience a streamlined onboarding flow designed to reduce setup friction and accelerate activation. This includes clearer onboarding progress visibility, simplified setup guidance, and stronger prompts encouraging users to complete their first tracked ride.

Both groups will receive the same lifecycle reminders during the experiment period, so the test isolates the impact of onboarding improvements themselves.

Primary Metric

First-Ride Activation Rate (7-day)

Percentage of users who complete first_ride_tracked within 7 days of signup.

Activation Rate = Users who complete first_ride_tracked within 7 days ÷ Total new users

NOTE: A 7-day window was selected because Quin is a real-world riding product, and ride opportunities may vary depending on commuting habits, weather, or weekend riding behavior.

Guardrail Metrics

To ensure onboarding simplification does not negatively impact product quality or safety readiness, I would monitor several guardrail metrics alongside activation performance. These include device pairing completion, emergency contact setup completion, onboarding completion rate, app uninstall rate, and onboarding-related application errors or crashes.

Minimum Detectable Effect (MDE)

To estimate the baseline activation rate used in the experiment design, I analyzed the proportion of newly registered users who completed first_ride_tracked within 7 days of signup. Using the provided event data, 672 out of 2,000 users completed a first tracked ride during their first week, resulting in a baseline activation rate of approximately 33.6%.

Code reference - https://colab.research.google.com/drive/1KwVZDGZANY33Eh1NgPpsM6hnymvRwtHe#scrollTo=FVnV1h_93IxT&line=1&uniqifier=1

The experiment will target an absolute improvement of +5 percentage points, increasing the conversion rate from 33.6% to 38.6%. This corresponds to an expected relative lift of approximately 15%, which would represent a meaningful improvement in early activation performance.

Sample Size & Estimated Duration

Using a significance level (α) of 0.05, statistical power of 80%, and a target lift of approximately 5 percentage points, the experiment would likely require a few thousand users in total to reliably detect a meaningful improvement in activation performance.

Assuming Quin acquires approximately 200 eligible new users per day with a 50/50 experiment split, the estimated experiment runtime would be roughly 2–3 weeks. Because the primary metric measures 7-day activation, the analysis window would include an additional 7 days after enrollment closes to allow all users to complete the full conversion period before evaluation.

Instrumentation & Data Requirements

To ensure valid experimentation and reliable attribution, we would need consistent onboarding, activation, and lifecycle event tracking across both variants. This includes onboarding progression events, device pairing events, ride-tracking activity, onboarding completion timestamps, and experiment metadata such as experiment_id, variant, platform, device type, and app version.

Validity Considerations

Several safeguards should be implemented to ensure the experiment results are trustworthy and unbiased. Internal and test accounts should be excluded from analysis, and randomization balance across iOS and Android users should be verified before evaluation. Event logging should be validated prior to launch to ensure onboarding and activation events are consistent across both variants. Sample ratio mismatch monitoring would also be important to ensure traffic allocation remains consistent with the intended experiment split.

Secondary analysis will segment users by device pairing completion to distinguish onboarding friction from potential hardware-readiness constraints.

Success Criteria

The experiment would be considered successful if the treatment group shows a statistically significant increase in first_ride_tracked conversion while maintaining stable onboarding quality and safety-related guardrail metrics.

Task 3 - A lifecycle message you would send

1. Segment Identification & Logic

For this lifecycle intervention, I focused on premium trial users who showed early signs of disengagement after starting their trial. Specifically, I identified users who triggered premium_trial_started but had not yet converted to premium and showed no recent app or ride-tracking activity during the last 7 days of the dataset.

This segment was selected because starting a premium trial indicates high initial intent, while prolonged inactivity suggests these users may not be experiencing enough ongoing value to continue engaging with the product.

How the Segment Was Sized

Users were classified as "at-risk" if they:
• triggered premium_trial_started,
• had not triggered premium_converted,
• had not triggered subscription_cancelled,
• and showed no recent app or ride-tracking activity for at least 7 days.

Using this logic, I identified 70 at-risk users, representing approximately 36% of all premium trial users in the dataset.

Code link for this section –

https://colab.research.google.com/drive/1KwVZDGZANY33Eh1NgPpsM6hnymvRwtHe#scrollTo=eQ2TXnbva_Yp&line=2&uniqifier=1

Message to be sent and Channel:

Since this segment consists of premium trial users who already demonstrated high purchase intent but are now becoming inactive, I would use an email-based lifecycle intervention rather than a push notification. A 7-day inactivity period during a 14-day premium trial suggests meaningful disengagement, while email provides enough space to reinforce Quin's core safety value proposition and encourage users to return before the trial expires.

SUBJECT - Ready for your next ride? Let's get you protected.

PREVIEW TEXT - Your Quin Premium trial is active. Make sure your next ride is fully protected.

Hi {first_name},

We noticed you haven't tracked a ride in a few days. Whether you've been busy or just haven't had a chance to ride recently, we want to make sure your next trip is fully protected.

Your Quin Premium trial is still active, which means you have full access to Quin's core safety features:

- Automatic crash detection
- Emergency alerts
- Premium ride tracking and safety insights

Your setup is already complete — you're ready to start your next ride with Quin.

[Track Your Next Ride]

Success Evaluation & Measurement

The primary success metric for this intervention would be the reactivation rate, defined as the percentage of targeted users who return and complete a ride_tracked event within 7 days of receiving the email. Secondary metrics would include premium conversion rate, email open rate, click-through rate, and downstream 30-day retention. To measure the true impact of the intervention, performance would be compared against a randomized control group of similar inactive premium trial users who did not receive the email.

Task 4 - Predictive Model Sketch

For this exercise, I approached the problem as an early engagement prediction task using Quin's onboarding and behavioral event data. Because the dataset only covers a 90-day activity window with users signing up across a compressed acquisition period, a traditional churn model based purely on uninstall or cancellation behavior would be incomplete and noisy.

Instead, I framed the problem as:

"Can we predict, from a user's first two weeks of behavior, whether they are likely to remain engaged during the following weeks?"

This creates a more actionable setup for lifecycle and retention interventions.

Target Variable & Labeling Logic

Target Variable: is_retained_day_15_to_30

This is a binary target variable:

- **1 = Retained / Engaged**
- **0 = Not Retained**

Labeling Logic: A user was labeled as Retained (1) if they triggered at least one app_opened event between Days 15–30 after signup.

Users were labeled as Not Retained (0) if they showed no return app activity during this period or triggered hard churn signals such as app_uninstall or subscription_cancelled before Day 30**.**

Feature Engineering Choices

I created features using only the user's first 14 days of activity so the model could learn from early behavior without leaking future information.

The features mainly focused on four areas:

- onboarding completion,
- engagement intensity,
- core product adoption,
- and acquisition quality.

For onboarding, I looked at whether users:

- paired their device,
- added an emergency contact,
- and how quickly they completed these steps after signup.

For engagement behavior, I included:

- number of app opens,
- active days,
- total event count,
- and ride tracking activity in the first two weeks.

I also added features tied to Quin's core product value, such as:

- ride_tracked,
- first_ride_tracked,
- sos_test_triggered,
- and whether the user started a premium trial.

Finally, I included user-level attributes like:

- acquisition source,
- device model,
- country,
- and plan type.

Baseline Model & Performance

I trained a Random Forest classifier using behavioral and onboarding features from the user's first 14 days after signup. Link here – https://colab.research.google.com/drive/1KwVZDGZANY33Eh1NgPpsM6hnymvRwtHe#scrollTo=miUQunwzC0sO&line=341&uniqifier=1

The model was evaluated using a stratified train/test split and standard classification metrics.

Metric	**Score**
ROC-AUC	0.967
Precision	0.978
Recall	0.771
F1 Score	0.863

Interpretation

The baseline model showed strong predictive performance within the limited observation window, suggesting that early onboarding and engagement behavior contains useful signals about future user retention.

The most important predictive features were:

- early ride tracking activity,
- number of active days,
- device pairing completion,
- and overall app engagement frequency.

The model achieved high precision, meaning it was generally accurate when identifying users likely to remain engaged, although recall was lower, suggesting there is still room to improve identification of at-risk users who eventually disengage.

What I Would Do Next

If this were a real production project, I would first validate the model across longer retention windows (30-day, 60-day, and 90-day retention) to ensure the predictive signal generalizes beyond the short observation window used here.

I would also experiment with additional modeling approaches and evaluate feature importance more deeply. The current dataset already contains strong behavioral signals, including onboarding events, notification engagement, ride activity, and monetization events. However, there are still several additional data points that would improve the model:

- richer ride telemetry beyond duration alone (ride consistency, riding frequency trends, time-of-day usage),
- Bluetooth/device connectivity reliability logs,
- app crash or performance events,
- Lifecycle campaign metadata (notification type, send timing, campaign variant),
- longer-term subscription lifecycle history beyond the initial trial period

For example, a user stopping ride tracking may reflect hardware or connectivity issues rather than true disengagement. Similarly, understanding which notification campaigns successfully re-engage users would improve both prediction accuracy and lifecycle targeting strategy.

Several disengagement behaviors may reflect technical or onboarding friction rather than true loss of user interest, so deeper instrumentation would likely improve both prediction accuracy and lifecycle targeting.

Task 5- Slack Message to Engineering Team

Slack Message –

Hi team,

I have been working on a churn/retention modelling exercise using the Users and Events datasets. Overall, the instrumentation coverage already looks quite strong, especially around app activity, ride tracking, notifications, subscription lifecycle events, and acquisition attribution.

I wanted to seek your thoughts on potentially tracking a few additional events/properties that could be helpful for future growth experiments and retention analysis.

1. **Device / sync failure instrumentation event:** device_sync_failed

Reason: If helmet/device sync reliability impacts user experience, this could become an important early churn signal.

2. **App stability instrumentation event:** app_crash_detected

Reason: Reason: Would help identify whether app crashes or technical issues are contributing to onboarding drop-off, failed ride completion, or lower long-term retention.

3. **Notification fatigue signals event:** notification_dismissed

Reason: We already track push_notification_received and push_notification_tapped events along with campaign IDs. This would help measure notification fatigue and understand when messaging starts negatively impacting engagement.

Additionally, if we can track:

- channel (push, in_app_message, email)
- variant_id (control, copy_v2, etc.)

for notification events, that would be very helpful in understanding which engagement strategies are working best with our users.

Would love to know which of these can be easily made available to track vs which would require additional implementation effort. Happy to incorporate accordingly based on feasibility and current platform limitations.